Girls Build Bots — Brownie Cybersecurity Basics

Badge-aligned workshop guide for robotics-team-led outreach

Level: Brownies (Grades 2–3) · **Badge:** Cybersecurity — Cybersecurity Basics **Series:** Cybersecurity (Basics · Safeguards · Investigator) **Format:** One 75–90 minute session · **Ratio:** 1 mentor per 3–4 Scouts **Who runs it:** A FIRST robotics team (FLL / FTC / FRC) leading the session. A physical robot is a nice bonus but **not required** — every activity runs on free browser games and simple props the team or troop can provide.

> **Troop Leaders:** This curriculum is designed to *align with* the Brownie Cybersecurity badge series. Official badge requirements are in your Girl Scout Volunteer Toolkit (VTK). Verify completed activities against the official steps before awarding badges. Badges are purchased through your council shop.

The through-line: being safe and smart online

The Cybersecurity series builds digital street-smarts: understand how information travels (**Basics**), protect it (**Safeguards**), and think critically to solve problems (**Investigator**). For Brownies, keep it concrete and friendly — never scary.

Vocabulary: information, private, network, message, safe, personal information.

Cybersecurity Basics (75–90 minutes)

Badge theme: Learning how computers and networks share information, and why protecting personal information matters online.

Materials

- Devices with **Interland** (Google's *Be Internet Awesome*, beinternetawesome.withgoogle.com) — the "Mindful Mountain" and "Kind Kingdom" realms fit best
- Yarn or a ball to pass, for the network game
- "Private vs. OK-to-share" sorting cards (favorite color = OK; home address = private)

Activity 1 — Pass the Message (Network Game) (15 min)

Scouts stand in a circle and pass a message ("the robot is blue") hand to hand — that's a **network**, how information travels between computers. Then add a "listener" in the middle who tries to overhear. **Point:** messages travel through many hands, so some information we keep private.

Activity 2 — Private or Share-Able? (15 min)

Sort cards into two piles: things that are fine to share (favorite animal, your team name) vs. **personal information** we keep private (full name + address, phone number, passwords). Mentors guide the tricky ones. This is the core Basics idea — knowing the difference.

Activity 3 — Play Interland (35 min)

Scouts play the friendly, age-appropriate Interland game (mentor-led, one realm). It reinforces sharing wisely and being kind online, through play. Mentors circulate and connect what happens in the game back to the sorting activity.

Activity 4 — My Three Rules (5 min) + Wrap (5 min)

Each Scout names one thing she'll keep private and one thing that's fine to share. Closing circle: thumbs up if you can spot personal information now.

> **Robot moment (optional):** Share for 60 seconds how the team keeps its robot code and passwords private so no other team can change their robot — being careful with information matters in robotics too.

Completion Checklist — Cybersecurity Basics

By the end of the session, to earn the badge, each Scout can: - [ ] Explain that a network is how information travels between computers - [ ] Tell the difference between personal information and things that are OK to share - [ ] Play through an online-safety activity and name what it taught - [ ] Name at least one piece of personal information she'll keep private

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Mentor Guide (for the robotics team)

- **You don't need a robot for this.** Free browser games and a ball of yarn are enough. The "robot moment" box is optional.
- **Keep it friendly, never scary.** The goal is confidence and smart habits, not fear of the internet.
- **Make "private" concrete.** Use examples from their world (name of their school, home address) rather than abstract warnings.
- **Model curiosity, not lecturing.** Ask "would you share this? why or why not?" and let them reason.
- **Watch the clock.** Brownies fade after ~25 minutes; keep segments moving.

📣 Share Your Impact — required for teams and troops using this curriculum

If you use Girls Build Bots materials to run a workshop, please complete all three:

1. **Credit us.** Tell your Scouts and audience that the curriculum comes from Girls Build Bots, and credit "Girls Build Bots" anywhere you share about the event.
2. **Tag us.** Post about your workshop and tag **@girlsbuildbots** on Instagram.
3. **Report your results.** Email **girlsbuildbots@gmail.com** with: the number of girls who participated, the number of badges earned, and a sentence or two of feedback on how it went.

This is how the program grows and how we measure our impact — your report becomes part of the Girls Build Bots story and helps the next troop and team.

Created for the "Girls Build Bots" Girl Scout Gold Award project. Original curriculum aligned to the Brownie Cybersecurity badge series; official requirements © Girl Scouts of the USA, available to troop leaders in the Volunteer Toolkit.